
Real-Time Analytics Dashboard

Machine Learning Methods for Data Streams

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1 Introduction

The project introduces a `StreamDashboard` class - a general-purpose, real-time 8-panel analytics dashboard inspired by the taxonomy of Webb et al. (2018) - that simultaneously monitors classification performance, feature importance, prediction confidence, feature-level drift, and prior (label) drift. Four distinct data streams are evaluated: the synthetic SEA generator with injected abrupt drift, the IoT Occupancy Detection real-world dataset, the Hyperplane generator with continuous drift, and a real-world weather dataset with gradual concept change. All experiments use the Hoeffding Adaptive Tree classifier as the primary learner, with ADWIN as the drift detector for the real-world streams. Results demonstrate that the dashboard effectively reveals multiple manifestations of concept drift across all stream types, and that HAT adapts its internal structure in response to the detected changes.

This project addresses the following tasks:

- Selection of two real-world and two synthetic data streams.
- Development of a novel multi-panel stream analytics dashboard (**`StreamDashboard`**).
- Evaluation of the HAT classifier and the dashboard across all four streams.

2 Data Streams

2.1 Synthetic Streams

2.1.1 SEA Generator with Injected Abrupt Drift

The SEA concepts [3] are among the most frequently used benchmarks for concept drift research. Each sample consists of three features $f_1, f_2, f_3 \in [0, 10]$, where f_3 is irrelevant noise. Classification is determined by a threshold on $f_1 + f_2$:

$$y = \begin{cases} 1 & \text{if } f_1 + f_2 \leq \theta \\ 0 & \text{otherwise} \end{cases} \quad (1)$$

In this project, SEA variant 0 uses $\theta = 8$ (before drift), and variant 2 uses $\theta = 7$ (after drift). Abrupt drift is injected at step 1500 out of 3000 total samples. This change shifts the decision boundary inward, causing an immediate decrease in classification accuracy for any model that has adapted to the previous concept.

2.1.2 Hyperplane Generator with Continuous Drift

The Hyperplane generator [4] produces a 10-dimensional binary classification problem defined by a time-varying hyperplane:

$$y = \text{sgn}\left(\sum_{i=1}^d w_i(t) \cdot x_i - w_0\right) \quad (2)$$

where the weight vector $\mathbf{w}(t)$ rotates continuously at each time step with magnitude `mag_change` = 0.01. Five out of ten features drift, making this a challenging gradual covariate-shift scenario. The stream is initialised with `seed=42`, runs for 3000 samples, and uses 10 features with 5 drifting dimensions.

2.2 Real-World Streams

2.2.1 Occupancy Detection (IoT Sensor Data)

This dataset [5] consists of sensor readings from a single office room used to predict binary room occupancy. The stream contains 8143 samples across five features:

Table 1: Occupancy Detection dataset features

Feature	Type	Description
Temperature	Continuous	Ambient temperature (°C)
Humidity	Continuous	Relative humidity (%)
Light	Continuous	Light intensity (Lux)
CO2	Continuous	CO ₂ concentration (ppm)
HumidityRatio	Continuous	Derived humidity ratio
Occupancy	Binary	Ground truth label (0 = empty, 1 = occupied)

The first 4000 samples are used, corresponding to roughly 1.5 days of measurements, a period during which the occupancy pattern (and thus the label distribution) changes naturally as the building transitions from working hours to night-time.

2.2.2 Weather Dataset

This publicly available weather dataset [6] contains approximately 18000 meteorological observations with 10 features (including temperature, pressure, wind speed and direction, dew point, etc.) and a binary target indicating whether it rained the following day. The stream is known to exhibit gradual concept drift driven by seasonal patterns. All samples are processed.

3 StreamDashboard

3.1 Motivation and Design Philosophy

Webb et al. [1] provide a comprehensive taxonomy of concept drift types and argue that effective stream mining tools must be capable of detecting multiple concurrent drift manifestations. Existing tools in `river`, `scikit-multiflow`, and related libraries focus on single-aspect monitoring (accuracy curves or detector alerts), leaving practitioners without an integrated, holistic view of stream behaviour.

The **StreamDashboard** addresses this gap by providing a self-updating, 8-panel visualisation that runs synchronously with the online learning loop. Its design goals are:

- **Generality:** Works with any `river` classifier and any number of features.
- **Interpretability:** Identifies which features drive classification and how they drift.
- **Flexibility:** Supports Accuracy, Cohen’s κ , and macro-averaged F1 as performance metrics.

3.2 Dashboard Panels

Table 2 summarises all eight panels and the drift type each monitors.

Table 2: StreamDashboard panels and their monitoring targets

Panel	Title	Drift type monitored	Method
1*	Decision boundary	Concept drift	Contour plot on top-2 features
2*	Error heatmap	Spatial error distribution	2-D histogram of prediction errors
3*	Feature importance	Feature drift	$ w_j $ from auxiliary logistic regression
4*	Performance metric	Overall concept drift	Prequential accuracy / κ / F1
5	Rolling accuracy per class	Class-conditional drift	Rolling 100-sample accuracy
6	Confidence margin	Model uncertainty	EMA of $ P(1) - P(0) $
7	Feature drift heatmap	Covariate shift	σ -deviation (Eq. 5)
8	Prior distribution	Prior drift	Reference vs. current class proportions

* Panels utilizing the auxiliary logistic regression model: Panels 1 and 2 use it to dynamically select the top-2 features for their axes, Panel 3 visualizes its learned weights, and Panel 4 tracks its performance as a baseline.

3.2.1 Top-2 Feature Selection (Panels 1 and 2)

At each visualisation step, the two features with the highest absolute weight in the auxiliary logistic regression model are selected as the axes for the decision boundary and error heatmap plots. Because the logistic model is updated online, this selection evolves dynamically as the stream progresses, providing an adaptive view of the most discriminative subspace.

All remaining features are frozen to their current window means when evaluating the decision boundary grid, ensuring a valid marginalisation over the remaining feature subspace.

3.2.2 Feature Importance (Panel 3)

To monitor feature-level drift, the dashboard visualises the dynamic importance of each feature. This is achieved by extracting the absolute weights, $|w_j|$, from the auxiliary logistic regression pipeline running in parallel with the main classifier. Because this auxiliary model is updated online, the weights reflect the current discriminative power of each feature. Significant changes in the relative ordering or magnitude of these weights over time serve as a strong indicator of covariate shift or feature drift, allowing practitioners to identify exactly which input variables are driving the model’s decisions at any given moment.

3.2.3 Performance Metric (Panel 4)

Panel 4 provides a macroscopic view of the model’s overall health by tracking continuous performance metrics, which is the standard approach for detecting overall concept drift. Depending on the user’s configuration and the nature of the data stream (e.g., class imbalance), this panel plots the prequential accuracy, Cohen’s κ , or macro-averaged F1 score. A sudden, sharp degradation in this metric typically signals abrupt concept drift, whereas a slow, sustained decline may point towards gradual drift that the model’s internal adaptation mechanisms are failing to accommodate.

3.2.4 Rolling Accuracy per Class (Panel 5)

While the overall performance metric provides a global view, it can mask asymmetric degradation, especially in data streams with skewed label distributions. To address this, Panel 5 monitors class-conditional drift by calculating the rolling accuracy over the last 100 samples independently for each class. This granular view makes it immediately evident if the classifier is maintaining high accuracy on the majority class while completely failing on the minority class—a common pathology during complex concept drift that a single global metric might otherwise obscure.

3.2.5 Prediction Confidence Margin (Panel 6)

The confidence margin at step t is computed as the absolute difference between class posterior probabilities output by the main model:

$$m(t) = |P(\hat{y} = 1 \mid x_t) - P(\hat{y} = 0 \mid x_t)| \quad (3)$$

To reduce noise, an exponential moving average (EMA) with $\alpha = 0.1$ is applied:

$$\tilde{m}(t) = 0.9 \cdot \tilde{m}(t - 1) + 0.1 \cdot m(t) \quad (4)$$

A sustained decrease in $\tilde{m}(t)$ indicates that the model is becoming uncertain about new samples - a reliable early signal of concept drift, often preceding a visible drop in accuracy.

3.2.6 Window-Based σ -Deviation Feature Drift (Panel 7)

The StreamDashboard implements a custom per-feature deviation monitor. At each step t (sampled every 10 steps), two windows are maintained:

- **Current window:** the most recent `window_size` samples.

- **Reference window:** a window of the same size ending `ref_lag` steps ago.

For each feature j , the standardised deviation is computed as:

$$\delta_j(t) = \text{clip}\left(\frac{\bar{x}_j^{\text{cur}}(t) - \bar{x}_j^{\text{ref}}(t)}{\sigma_j^{\text{ref}}(t) + \varepsilon}, -3, 3\right) \quad (5)$$

where $\bar{x}_j^{\text{cur}}$ and $\bar{x}_j^{\text{ref}}$ are the feature means in each window, σ_j^{ref} is the standard deviation in the reference window, and $\varepsilon = 10^{-9}$ prevents division by zero. The clipping to $[-3\sigma, +3\sigma]$ removes outlier influence. The matrix $\{\delta_j(t)\}_{j,t}$ is rendered as a heatmap, where red indicates a positive (increasing) shift and blue indicates a negative (decreasing) shift.

3.2.7 Prior Drift Monitor (Panel 8)

Plot 8 of the dashboard compares the class distribution (prior $P(y)$) in the reference and current windows. Formally, it computes:

$$\Delta P(y = c) = P_{\text{cur}}(y = c) - P_{\text{ref}}(y = c), \quad c \in \{0, 1\} \quad (6)$$

A visually significant difference in the bar chart indicates label distribution shift, which is a necessary (but not sufficient) condition for concept drift.

4 Experiments

4.1 Setup

All experiments use the Hoeffding Adaptive Tree as the main classifier. Table 3 summarises the configuration for all four experiments.

Table 3: Experimental configuration

Stream	Samples	Window	Ref lag	Update freq	Metric
SEA (synthetic)	3000	300	500	100	Accuracy
Occupancy (real)	4000	300	500	150	Cohen’s κ
Hyperplane (synth.)	20000	300	500	100	Cohen’s κ
Weather (real)	18159	500	1000	100	Cohen’s κ

4.2 Experiment 1: SEA Generator with Abrupt Drift

The SEA experiment provides a controlled setting where the ground truth drift location (step 1500) is known.

In Figure 1 Panel 3 (Feature importance) confirms that the auxiliary logistic regression correctly relies exclusively on the relevant features (f_1 and f_2), effectively assigning near-zero weight to the noise feature (f_3). Consequently, Panel 1 visualizes HAT’s step-like spatial approximation of the true linear decision boundary based solely on this highly discriminative top-2 feature subspace. The most critical observation from the dashboard

is the manifestation of the abrupt concept drift injected at step 1500. While the global prequential accuracy (Panel 4) remains high (0.94) and entirely obscures the drift due to its cumulative nature, Panel 5 (Rolling accuracy per class) explicitly captures the event. Following the boundary shift (from 8to7), the rolling accuracy for Class 0 abruptly plateaus at 100%, accompanied by a simultaneous degradation and increased variance in Class 1 accuracy. This perfectly illustrates the asymmetric nature of this specific threshold change, validating the necessity of class-conditional monitoring. Additionally, the dashboard successfully distinguishes the type of drift occurring in the stream. The feature drift heatmap (Panel 7) shows minimal deviations ($\approx 0\sigma$) across all features, and the prior distribution (Panel 8) remains stable between the reference and current windows. This combination correctly verifies that the injected SEA drift is a pure concept drift without any underlying covariate shift or label distribution skew. Finally, the confidence margin (Panel 6) exhibits heightened volatility and slight degradation leading up to and following the drift point, demonstrating its value as a supplementary indicator of model uncertainty during spatial restructuring.

The external ADWIN detector complemented these metrics by signaling a statistically significant change at step 895; however, achieving this detection required an increased sensitivity setting of $\delta = 0.5$, highlighting the inherent trade-off between prompt drift signaling and the suppression of false positives in synthetic environments.

4.3 Experiment 2: Occupancy Detection

The evaluation of the real-world IoT Occupancy dataset provides a stark contrast to the synthetic SEA stream, serving as a prime example of complex, multi-faceted drift driven by natural temporal cycles (e.g., day-to-night transitions).

In Figure 2 we can see that at step 3900, the dashboard effectively exposes the anatomical differences between artificial concept drift and real-world environmental shifts. Panel 3 (Feature importance) aligns perfectly with domain intuition, identifying Light as the dominant predictive feature, closely followed by CO₂ levels. This confirms that the auxiliary logistic regression correctly captures the physical markers of human presence. The most striking visual evidence of real-world dynamics is found in Panel 7 (Feature drift heatmap). Unlike the purely white heatmaps of the synthetic stream, Panel 7 displays massive, synchronized vertical bands of deep red ($+3\sigma$) and deep blue (-3σ) across all sensor readings. These bands represent severe Covariate Shift, corresponding to the office transitioning between occupied working hours (temperatures, light, and CO₂ rising) and unoccupied night hours (values dropping). Accompanying this is an extreme Prior Drift, explicitly confirmed by Panel 8, which shows that at step 3900, the class distribution has collapsed entirely to 100% Class 0 (unoccupied). Despite these massive environmental shifts, the Hoeffding Adaptive Tree maintains a remarkably high overall prequential κ of 0.920 (Panel 4). However, the dashboard’s diagnostic panels reveal the model’s localized struggles during the day/night transitions. Panel 6 (Prediction confidence margin) exhibits distinct, deep valleys that perfectly align with the transition phases seen in the covariate heatmap. Furthermore, Panel 5 (Rolling accuracy per class) shows that while Class 0 remains highly stable, Class 1 accuracy becomes highly erratic and periodically vanishes a direct consequence of the minority class completely disappearing from the sliding window during night-time. Finally, the spatial error distribution (Panel 2) confirms that the model’s mistakes are highly concentrated in the "twilight" transition zones (moderate light and

CO₂ levels), rather than during the easily predictable dead of night.

Ultimately, this experiment validates the dashboard’s ability to simultaneously monitor Covariate Shift, Prior Drift, and transition-induced Concept Drift, providing a holistic view of a classifier operating in a highly cyclical physical environment.

The robustness of this monitoring is further corroborated by the ADWIN detector, which signaled significant concept drift at steps 1152, 1312, 3840, and 3936. These detections align precisely with the ‘twilight’ transition phases, confirming that the model effectively recognizes and reacts to the shifting environmental correlations as the office moves between its distinct operational states.

4.4 Experiment 3: Hyperplane Generator

The evaluation of the Hyperplane generator provides a contrasting scenario to the SEA experiment, specifically demonstrating the dashboard’s ability to monitor continuous, gradual concept drift. Driven by the `mag_change=0.01` parameter across 5 of the 10 dimensions, the underlying decision boundary rotates persistently throughout the 20000-step stream.

As shown in Figure 3 this continuous evolution is most evident in the global performance metric (Panel 4). Unlike the abrupt drop seen in sudden drift scenarios, Cohen’s κ peaks early around 0.70 and then exhibits a slow, sustained erosion, ending at 0.64. This trajectory is characteristic of gradual drift, illustrating the HAT continuous struggle to keep its internal leaf statistics aligned with a moving target. Correspondingly, Panel 5 (Rolling accuracy per class) shows highly intertwined, noisy fluctuations for both classes. Because the hyperplane rotates smoothly, the degradation does not disproportionately target a single class (unlike the asymmetric threshold shift in SEA), resulting in relatively symmetric instability. The dashboard also effectively captures the high-dimensional nature of the problem. Panel 3 (Feature importance) reveals a distributed reliance on multiple features, with the auxiliary logistic regression currently identifying features 1 and 4 as the most discriminative (top-2) at step 20000. Furthermore, Panel 6 (Prediction confidence margin) displays sustained, high volatility and generally lower confidence levels (often dropping below 0.5). This reflects the model’s persistent uncertainty in a non-stationary 10-dimensional space where the concept never fully settles. Finally, Panels 7 and 8 confirm the structural nature of the Hyperplane stream. The class priors (Panel 8) remain perfectly stable near a 50/50 split, and the feature drift heatmap (Panel 7) shows negligible deviation ($\approx 0\sigma$).

This combination confirms that the continuous degradation in κ is driven entirely by Real Concept Drift (the shifting relationship $P(y|X)$ dictated by the rotating weights), rather than Covariate Shift in the underlying uniform input space.

The statistical confirmation of this process was provided by the ADWIN detector, which signaled a concept drift at step 2999. This late-stage detection highlights the inherent latency of threshold-based monitoring in continuous drift environments, where the cumulative impact on the error stream only reaches the level of statistical significance after a prolonged and sustained erosion of the model’s predictive power.

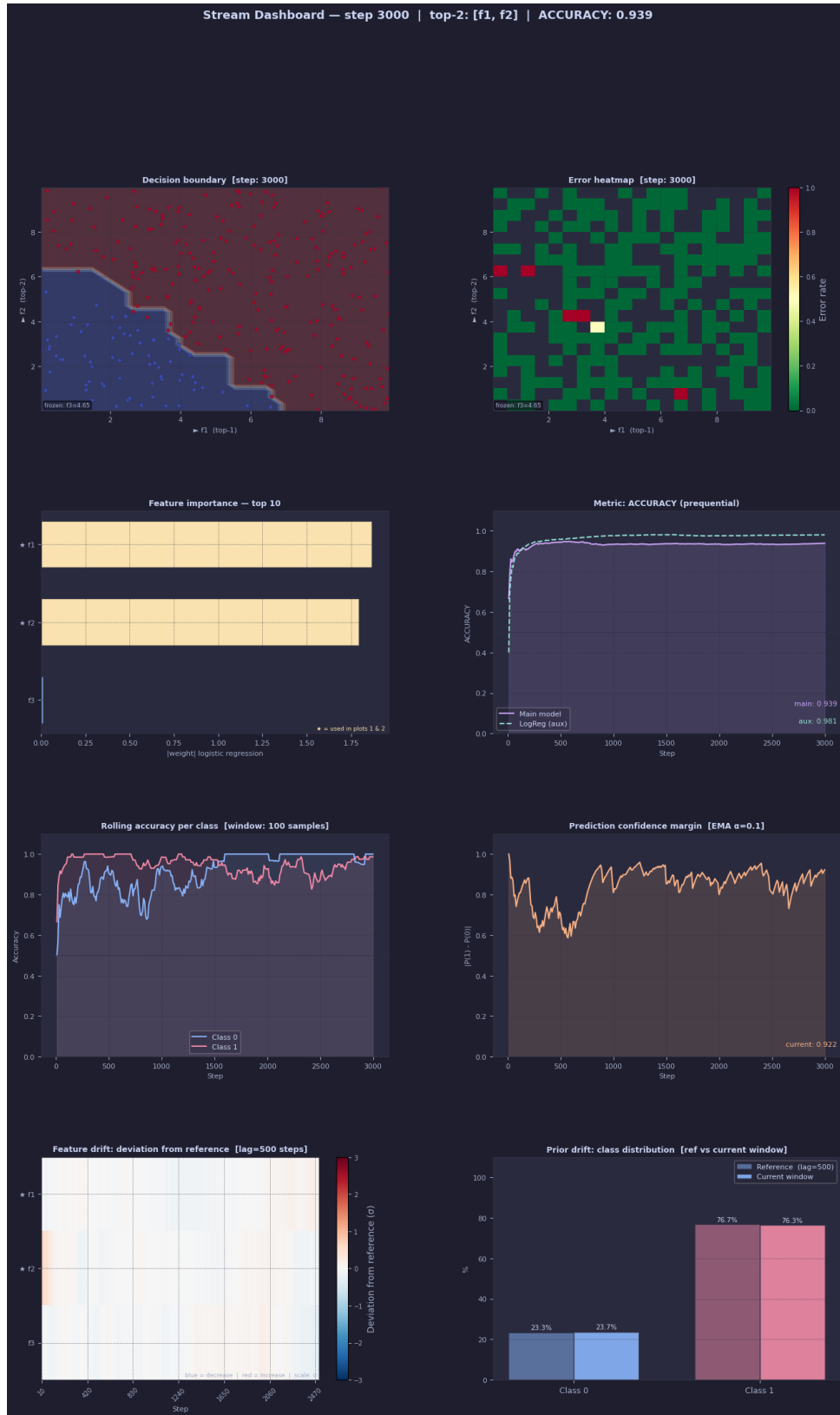


Figure 1: Experiment 1 - Dashboard



Figure 2: Experiment 2 - Dashboard

4.5 Experiment 4: Weather Dataset

The evaluation of the real-world Weather dataset represents the most challenging and noisy environment among the four tested streams. Captured at step 18100, the dashboard illustrates a scenario dominated by high inherent uncertainty and complex, overlapping distributions, typical of long-term meteorological forecasting.

The most prominent feature of this experiment is the inherently low predictive performance. In Figure 4 Panel 4 (Metric: κ) shows that after an initial learning phase, the prequential Cohen’s κ plateaus at a remarkably low level, ending at 0.36. This indicates that predicting rain based on the available features is exceptionally difficult. This difficulty is corroborated by Panel 6 (Prediction confidence margin), which remains highly volatile and depressed (often fluctuating between 0.4 and 0.6), demonstrating that the HAT rarely achieves high certainty in its predictions. Panel 2 (Error heatmap) and Panel 1 (Decision boundary) provide deep geometric insights into why the model struggles. The data points mapped onto the top-2 discriminative features (feat_7 and feat_2) form a tight diagonal band, implying a strong correlation between these variables. More importantly, the errors (yellow and red blocks in Panel 2) are heavily dispersed throughout the entire dense region of the data space rather than being isolated near a clear decision boundary. This suggests a high Bayes error rate—the classes are fundamentally overlapping in this subspace.

Furthermore, the dashboard captures the erratic nature of the model’s short-term performance. Panel 5 (Rolling accuracy per class) is chaotic, with both the majority class (Class 0, 65% prior) and the minority class (Class 1, 35% prior) suffering from violent accuracy swings. Unlike the massive, synchronized cyclical shifts seen in the Occupancy dataset, the drift in the Weather dataset is far more subtle and localized. Panel 7 (Feature drift heatmap) reveals spikes of covariate shift (e.g., sudden deviations in feat_3 and feat_4 at earlier steps), likely corresponding to specific seasonal changes or short-term weather anomalies.

This erratic behavior is statistically substantiated by the ADWIN detector, which flagged multiple concept drifts at steps 1344, 1728, 2080, 3328, 13344, 16960 and 18048. These frequent triggers reflect the non-stationary and highly volatile nature of the meteorological features, confirming that the high Bayes error and fundamentally overlapping class distributions necessitate constant recalibration of the model’s internal decision logic.

4.6 Dashboard as a Novel Evaluation Tool

Beyond standard numeric metrics, the `StreamDashboard` contributes to evaluation in several non-trivial ways:

5 Code and Data Sources

5.1 Code Repository

The complete source code for this project, including the implementation of the `StreamDashboard` class, data stream configurations, and scripts required to reproduce all experiments presented in this report, is publicly available on GitHub. The repository can be accessed at the following URL:



Figure 3: Experiment 3 - Dashboard

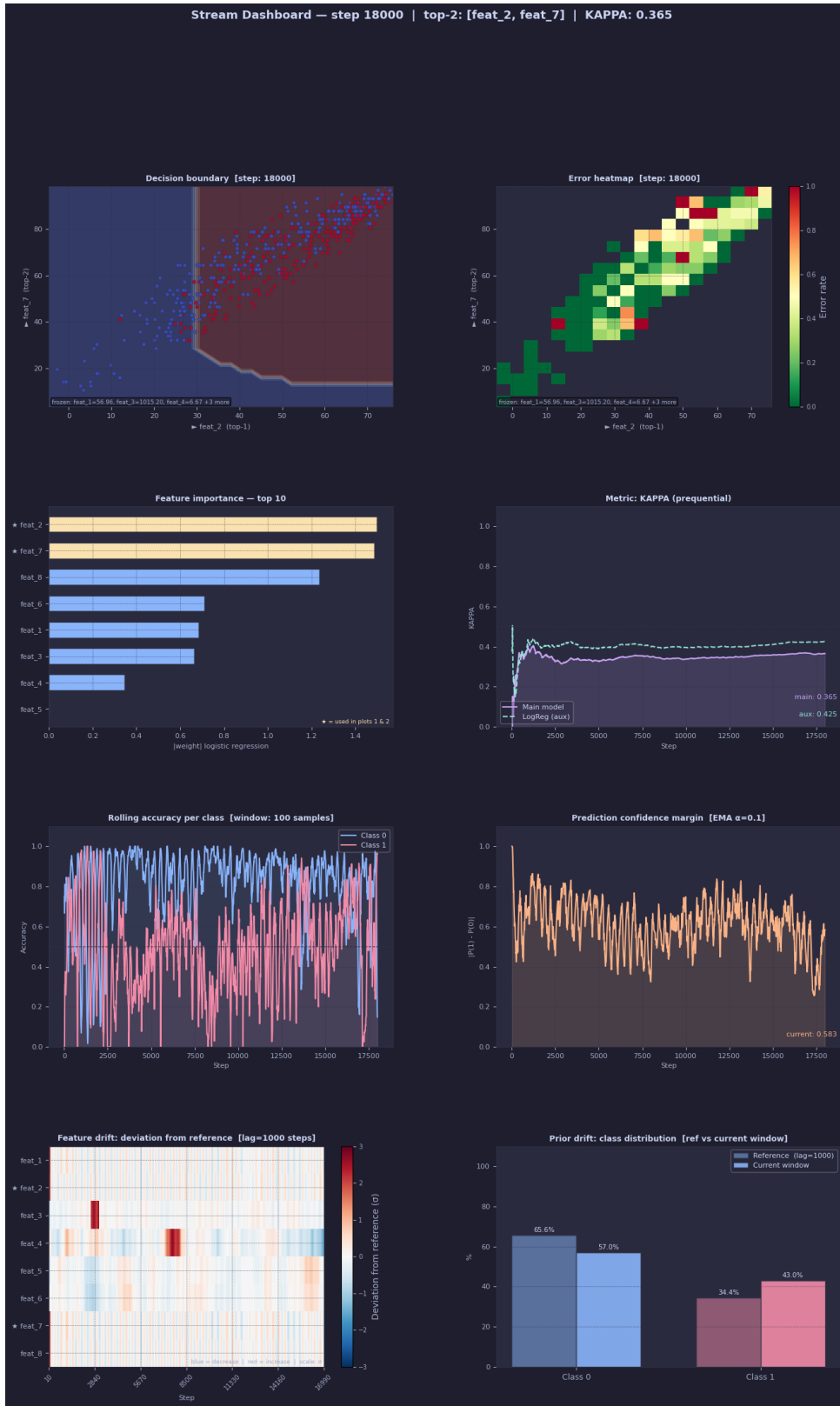


Figure 4: Experiment 4 - Dashboard

<https://github.com/Bartekk17/data-streams-dashboard.git>

5.2 Data Sources

Table 4: Data source URLs

Dataset	Source
Occupancy Detection	https://github.com/LuisM78/Occupancy-detection-data
Weather	https://github.com/ogozuacik/concept-drift-datasets-scikit-multiflow
SEA Generator	<code>river.datasets.synth.SEA</code> (built-in)
Hyperplane Generator	<code>river.datasets.synth.Hyperplane</code> (built-in)

6 Discussion

6.1 Strengths of the Approach

The **StreamDashboard** successfully unifies multiple drift detection signals - typically scattered across separate tools and papers - into a single, coherent view. The auxiliary logistic regression model decouples feature importance estimation from the main classifier’s architecture, ensuring that the dashboard works equally well with tree-based, linear, and ensemble online learners. The use of standardised σ -deviation for feature-level drift removes the need for calibrating absolute thresholds per feature, making the method scale-invariant.

6.2 Limitations

1. **Binary classification only:** The current implementation assumes a two-class problem. The prior drift histogram (panel 8), confidence margin (panel 6), and per-class rolling accuracy (panel 5) all assume $y \in \{0, 1\}$. Extension to multiclass requires replacing the confidence margin with entropy and generalising the histogram to a bar chart with C bars.
2. **Auxiliary model lag:** The logistic regression weights take several hundred samples to stabilise, meaning that feature importance estimates during the first window may be unreliable.

7 Conclusion

This project presented a stream mining study covering four data streams (two synthetic, two real-world) with a range of concept drift types: abrupt (SEA), continuous covariate shift (Hyperplane), and complex real-world drift (Occupancy, Weather). The Hoeffding Adaptive Tree classifier with ADWIN-based internal adaptation was used as the primary learner, complemented by ADWIN applied to the error stream.

The principal novel contribution is the **StreamDashboard** class, an 8-panel real-time visualisation tool that monitors classification performance, spatial error distribution, dynamic feature importance, model confidence, per-feature covariate drift, and label distribution drift in a single, unified interface.

Future work includes extension to multiclass streams.

8 Updates compared to the preliminary report

List of Implemented Changes and Updates

- Comment: "Hyperplane stream is quite limited (in terms of instance count) given the dimensionality of the data. Please extend the evaluation period."
- Action taken: The evaluation period for the Hyperplane stream has been extended to include 20 000 number of instances.
- Comment: "The report states that up to 20000 instances of a stream of 18 000 instances were used (weather data). Please fix."
- Action taken: The text now accurately reflects that the maximum number of instances used for the weather dataset aligns with its actual size (18 159 instances).
- Comment: "Please add the performance of the auxiliary logistic regression model to panel 4 and make it clear in the other panels whether they rely on logistic regression." and "Please clarify all relevant details, e.g. in which of the cases logistic regression and reference model is used etc."
- Action taken: The performance of the auxiliary Logistic Regression model is now plotted as a baseline alongside the main model in Panel 4. Additionally, the report explanations have been updated (including asterisks in the summary table) to clearly indicate which panels rely on the auxiliary model (Panels 1, 2, 3, and 4).
- Comment: "Prediction Confidence Margin - is this one based on logistic regression as well? Please clarify in report and GUI."
- Action taken: The title of Panel 6 has been updated to explicitly state "Prediction confidence margin (Main model)". The report now clearly explains that this margin is calculated using the posterior probabilities of the main classifier, not the auxiliary logistic regression.
- Comment: "Please cross-check the language of your report to eliminate editorial issues such as 'deviatio monitor'."
- Action taken: Performed a thorough proofreading of the report to fix typographical errors and editorial issues.
- Comment: "Please explicitly refer to figure numbers e.g. in Sect. 4.2."
- Action taken: Updated the text across the report to ensure all figures are explicitly referenced by their numbers.
- Comment: "Please make figures larger in your report."
- Action taken: Adjusted the formatting of the document. All figures and dashboard screenshots have been enlarged.

Pending work Due to time/ constraints, the following suggested extensions have not been included in the current iteration.

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- Pseudocode Implementation (algorithm2e): The addition of a formalized algorithm using algorithm2e to map out ADWIN operations and confidence margin calculations is pending.
 - Alternative Main Method (e.g., Adaptive Random Forest) Wrapper Architecture

References

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